

Accessible Language Learning Platform

Overview

The Accessible Language Learning Platform is a cloud-based application designed to support learners with cognitive, linguistic, or sensory learning disabilities such as dyslexia, ADHD, autism spectrum-related processing challenges, or auditory comprehension difficulties. The platform aims to provide an inclusive and adaptable learning environment for language acquisition.

Technologies Used

Frontend

The frontend of the application is built using modern web technologies, focusing on a responsive and accessible user experience.

- **React:** A JavaScript library for building user interfaces.
- **Vite:** A fast frontend build tool that provides a quick development experience.
- **Tailwind CSS:** A utility-first CSS framework for rapidly building custom designs.
- **i18next:** An internationalization framework for React applications, enabling multi-language support.
- **React Router DOM:** For declarative routing in React applications.

Backend

The backend is a Node.js application responsible for handling API requests, database interactions, and business logic.

- **Node.js:** A JavaScript runtime for server-side development.
- **Express.js:** A fast, unopinionated, minimalist web framework for Node.js.
- **Mongoose:** An ODM (Object Data Modeling) library for MongoDB and Node.js, providing a straightforward, schema-based solution to model application data.
- **MongoDB:** A NoSQL database used for storing application data.
- **CORS:** Middleware to enable Cross-Origin Resource Sharing.
- **Dotenv:** For loading environment variables from a `.env` file.
- **jsonwebtoken:** For implementing JSON Web Tokens for authentication.
- **bcryptjs:** For hashing passwords.

- **Nodemon:** A utility that monitors for changes in your source and automatically restarts your server.

Project Structure

The project is organized into `frontend` (implicitly the root, as per `vite.config.js` and `package.json`) and `backend` directories.

Plain Text

```
Accessible-Language-Learning-Platform-for-Learners-with-Learning-Disabilities/
├─ UML/ # UML diagrams and related documentation
├─ backend/ # Node.js/Express.js backend
│   ├─ models/ # Mongoose schemas
│   ├─ routes/ # API route definitions
│   ├─ check_courses.js
│   ├─ check_db.js
│   ├─ courses_log.txt
│   ├─ package.json
│   ├─ server.js # Main backend server file
│   └─ ...
├─ public/ # Public assets
├─ src/ # React frontend source code
│   ├─ __tests__/ # Test files
│   ├─ assets/ # Static assets (images, etc.)
│   ├─ components/ # Reusable React components
│   │   ├─ common/ # Common components
│   │   ├─ Layout.jsx # Main layout component with accessibility toggles
│   │   ├─ LetterTracing.jsx
│   │   ├─ MiniGames.jsx
│   │   ├─ PhonicsModule.jsx
│   │   ├─ ProtectedRoute.jsx
│   │   └─ RhymeLab.jsx
│   ├─ context/ # React Context API for global state management
│   │   └─ AccessibilityContext.jsx # Manages accessibility settings
│   ├─ data/ # Static data or mock data
│   ├─ locales/ # Internationalization files (e.g., en.json, es.json)
│   └─ pages/ # Page-level React components (views)
│       ├─ AdminDashboard.jsx
│       ├─ Dashboard.jsx
│       ├─ DyslexiaCenter.jsx # Accessibility features and settings
│       ├─ GuardianDashboard.jsx
│       ├─ Landing.jsx
│       ├─ LessonPlayer.jsx
│       ├─ Profile.jsx
│       ├─ Quiz.jsx
│       └─ Register.jsx
```

```
| | | └─ SpeechPractice.jsx
| | └─ services/ # API service calls or external integrations
| | └─ styles/ # Global styles or utility CSS
| | └─ utils/ # Utility functions
| | └─ App.css
| | └─ App.jsx # Main application component, defines routing
| | └─ i18n.js # i18next configuration
| | └─ index.css
| | └─ main.jsx # Entry point for React application
| | └─ setupTests.js
| └─ .gitignore
| └─ Documentation.pages
| └─ Modules Breakdown.pdf
| └─ README.md
| └─ auth.test.js
| └─ backend.test.js
| └─ eslint.config.js
| └─ gamification.test.js
| └─ index.html
| └─ package-lock.json
| └─ package.json # Frontend dependencies and scripts
| └─ phonics.test.js
| └─ vite.config.js
| └─ voice.test.js
```

Key Features and Functionality

- **User Authentication:** Secure login and registration for different user roles (student, guardian, admin).
- **Accessibility Features:** Includes a Dyslexia Center with toggles for dyslexic-friendly fonts, increased font size, screen reader activation, and theme switching (light/dark mode).
- **Language Learning Modules:** Various modules for language acquisition, such as Phonics & Letter Sounds, Letter Tracing, Memory Match, Picture Matching, Word Builder, and Rhyme Time Lab.
- **Quizzes:** Interactive quizzes, including voice-based and multiple-choice formats.
- **Dashboards:** Separate dashboards for Admin, Guardian, and regular users to track progress and manage settings.
- **Internationalization:** Support for multiple languages (English and Spanish are configured).
- **Speech Practice:** Functionality for speech-related exercises.

- **Gamification:** Integration of gamification elements (e.g., XP) to enhance engagement.

Setup and Local Development

Prerequisites

- Node.js (v18 or higher)
- npm (Node Package Manager)
- MongoDB instance (local or cloud-hosted)

Quick Start

1. Clone the repository:

Bash

```
git clone https://github.com/vishnuyerneni-maker/Accessible-Language-Learning-Platform-for-Learners-with-Learning-Disabilities
cd Accessible-Language-Learning-Platform-for-Learners-with-Learning-Disabilities
```

2. Install Frontend Dependencies:

Bash

```
npm install
```

3. Install Backend Dependencies:

Bash

```
cd backend
npm install
cd ..
```

4. Configure Environment Variables:

Create a `.env` file in the `backend` directory with your MongoDB URI and any other necessary environment variables (e.g., `PORT`, `JWT_SECRET`).

Example `backend/.env` :

Plain Text

```
MONGO_URI=mongodb://localhost:27017/language_learning
PORT=5000
```

```
JWT_SECRET=your_jwt_secret_key
```

5. Start the Backend Server:

Bash

```
cd backend  
npm start # or npm run dev for nodemon  
cd ..
```

6. Start the Frontend Application:

Bash

```
npm run dev
```

7. **Access the Application:** Open your browser and navigate to the URL provided by Vite (usually `http://localhost:5173`).

Testing

The project includes various test files for both frontend and backend components:

- `auth.test.js`
- `backend.test.js`
- `gamification.test.js`
- `phonics.test.js`
- `voice.test.js`

To run tests, refer to the `package.json` scripts for specific commands (e.g., `npm test`).

Contributors

- vishnuyerneni-maker
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- YerneniVishnuVardhan

License

(Information not explicitly available in the README.md or package.json files, typically found in a LICENSE file.)

References

- [Accessible Language Learning Platform GitHub Repository](#) [1]